

Java Programming Exam: Basic Level

Time Allowed: 90 mins

Question 1: Operators

Write a Java program to solve the following problem:

Given two integer variables, `a` and `b`, swap their values without using a third variable. Demonstrate this with `a = 5` and `b = 10`.

Question 2: For Loop

Write a Java program that uses a for loop to print the first 10 positive integers and their squares. The output should be in the following format:

```
1 -> 1
2 -> 4
...
10 -> 100
````
```

## Question 3: For Loop and Conditionals

Write a Java program that uses a for loop and conditionals to print out all the even numbers from 1 to 20, inclusive. Also, print a message 'This is an even number.' next to the even numbers.

## Question 4: Arrays

Declare and initialize an array of five integers in Java. Then, write a Java program to find the average of these integers.

## Question 5: Arrays and Loops

Given the following array: `int[] numbers = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10};`

Write a Java program to calculate the sum of all elements in the array using a for loop.

## Question 6:

Modify the program to skip the number 5 during summation.